

Network Security & Cryptography Assignment Solutions

Question 1: Shift Cipher Analysis [10 Marks]

Attack Type: Known Plaintext Attack

Solution Process:

1. Given: Ciphertext "CIW" corresponds to plaintext "yes"
2. Find the shift value:
 - $C \rightarrow y$: C is position 2, y is position 24, $\text{shift} = (2-24) \bmod 26 = 4$
 - $I \rightarrow e$: I is position 8, e is position 4, $\text{shift} = (8-4) \bmod 26 = 4$
 - $W \rightarrow s$: W is position 22, s is position 18, $\text{shift} = (22-18) \bmod 26 = 4$
3. The shift value is 4 (or equivalently, shift left by 22)
4. Decrypt "XVIEWVWI" using shift of 4:
 - $X \rightarrow T, V \rightarrow R, I \rightarrow E, E \rightarrow A, W \rightarrow S, V \rightarrow R, W \rightarrow S, I \rightarrow E$

Answer: The plaintext is "TREASURE"

Question 2: Vigenère Cipher with Keyword "HEALTH" [10 Marks]

Message: "Life is full of surprises" **Keyword:** "HEALTH"

Solution Process:

1. Remove spaces and convert to uppercase: "LIFEISFULLOFSUPRISES"
2. Repeat keyword to match message length: "HEALTHHEALTHHEALTHHE"
3. Apply Vigenère encryption formula: $C = (P + K) \bmod 26$

Character-by-character encryption:

- $L(11) + H(7) = S(18)$
- $I(8) + E(4) = M(12)$
- $F(5) + A(0) = F(5)$
- $E(4) + L(11) = P(15)$
- $I(8) + T(19) = B(1)$
- $S(18) + H(7) = Z(25)$

- $F(5) + H(7) = M(12)$
- $U(20) + E(4) = Y(24)$
- $L(11) + A(0) = L(11)$
- $L(11) + L(11) = W(22)$
- $O(14) + T(19) = H(7)$
- $F(5) + H(7) = M(12)$
- $S(18) + H(7) = Z(25)$
- $U(20) + E(4) = Y(24)$
- $P(15) + A(0) = P(15)$
- $R(17) + L(11) = C(2)$
- $I(8) + T(19) = B(1)$
- $S(18) + H(7) = Z(25)$
- $E(4) + H(7) = L(11)$
- $S(18) + E(4) = W(22)$

Answer: "SMFPBZMYLWHMZYPBZLW"

Question 3: Hill Cipher [10 Marks]

Note: The key matrix is not provided in the assignment document. I'll demonstrate the process with a standard 2×2 example.

Message: "We live in an insecure world" **Assuming 2×2 key matrix:**

$$K = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$$

Solution Process:

1. Remove spaces and convert: "WELIVEINANINSECUREWORLD"
2. Group into pairs: "WE LI VE IN AN IN SE CU RE WO RL D"
3. Add padding if needed: "WE LI VE IN AN IN SE CU RE WO RL DX"
4. Convert each pair to numbers and apply Hill cipher:
 - For each pair (a,b): multiply by key matrix mod 26

Example for first pair "WE":

- $W=22, E=4$
- $[3\ 2][22] = [3 \times 22 + 2 \times 4] = [74] \equiv [22] \pmod{26} = W$
- $[5\ 7][4] [5 \times 22 + 7 \times 4] [138] [8] I$

Answer: The complete ciphertext would depend on the actual key matrix provided.

Question 4: S-DES Encryption [20 Marks]

Data: 10101110 **K₁:** 01111010 **K₂:** 10110010

S-DES Process:

Step 1: Initial Permutation (IP)

Apply IP table to plaintext:

- Input: 10101110
- After IP: 01111010

Step 2: Round 1 with K₁

1. Divide into left and right halves:

- $L_0 = 0111$
- $R_0 = 1010$

2. Expansion/Permutation on R₀:

- $R_0 = 1010 \rightarrow EP(R_0) = 11010101$

3. XOR with K₁:

- $11010101 \oplus 01111010 = 10101111$

4. S-Box substitution:

- Left 4 bits (1010) $\rightarrow S_0 \rightarrow 01$
- Right 4 bits (1111) $\rightarrow S_1 \rightarrow 00$
- Result: 0100

5. P4 Permutation:

- $0100 \rightarrow 0010$

6. XOR with L₀:

- $L_1 = R_0 = 1010$
- $R_1 = L_0 \oplus P4(S\text{-boxes}) = 0111 \oplus 0010 = 0101$

Step 3: Round 2 with K_2

1. Expansion/Permutation on R_1 :

- $R_1 = 0101 \rightarrow EP(R_1) = 01010101$

2. XOR with K_2 :

- $01010101 \oplus 10110010 = 11100111$

3. S-Box substitution:

- Left 4 bits (1110) $\rightarrow S_0 \rightarrow 11$
- Right 4 bits (0111) $\rightarrow S_1 \rightarrow 10$
- Result: 1110

4. P4 Permutation:

- $1110 \rightarrow 1101$

5. XOR with L_1 :

- $L_2 = R_1 = 0101$
- $R_2 = L_1 \oplus P4(S\text{-boxes}) = 1010 \oplus 1101 = 0111$

Step 4: Final Permutation (IP^{-1})

- Combine: $L_2R_2 = 01010111$
- Apply IP^{-1} : $01010111 \rightarrow 01110101$

Answer: The secured text (ciphertext) is **01110101**

Summary

1. **Question 1:** Known Plaintext Attack, plaintext = "TREASURE"
2. **Question 2:** Vigenère ciphertext = "SMFPBZMYLWHMZYPBZLW"
3. **Question 3:** Hill cipher result depends on the provided key matrix
4. **Question 4:** S-DES ciphertext = "01110101"